

TANQIU JIANG

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RESEARCH INTERESTS

AI Safety and Trustworthy Machine Learning: safety and robustness of large language models (LLMs) and LLM-based agents, jailbreak attacks and defenses, adversarial robustness of vision-language models, agent security under long-horizon threats, and privacy-preserving generative modeling.

EDUCATION

Stony Brook University

Ph.D. in Computer Science Advisor: Dr. Ting Wang

Stony Brook, NY

Aug 2023 – Present

University of Rochester

Master of Science in Data Science

Rochester, NY

Aug 2020 – Dec 2021

Lehigh University

Bachelor of Science in Computer Engineering (Minor: Data Science)

Bethlehem, PA

Aug 2016 – May 2020

PUBLICATIONS

Peer-Reviewed Conference Papers

- [C1] **Jiang, T.**, Wang, Y., Liang, J., & Wang, T. (2026). AgentLAB: Benchmarking LLM Agents against Long-Horizon Attacks. In *International Conference on Machine Learning (ICML 2026)*. [pdf]
- [C2] **Jiang, T.**, Liang, J., Zhu, R., Zhou, J., Ma, F., & Wang, T. (2026). Dynamic Token Reweighting for Robust Vision-Language Models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)*. [pdf]
- [C3] Wang, Y., **Jiang, T.**, Liang, J., Fleming, C., & Wang, T. (2026). MAGE: Safeguarding LLM Agents against Long-Horizon Threats via Shadow Memory. In *ACM Conference on Computer and Communications Security (CCS 2026)*. [pdf]
- [C4] Liang, J., **Jiang, T.**, Wang, Y., Zhu, R., Ma, F., & Wang, T. (2026). AutoRAN: Weak-to-Strong Jailbreaking of Large Reasoning Models. In *Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. [pdf]
- [C5] Liang, J., Wang, Y., Li, C., Zhu, R., **Jiang, T.**, Gong, N., & Wang, T. (2026). GraphRAG Under Fire. In *IEEE Symposium on Security and Privacy (S&P 2026)*. [pdf]
- [C6] **Jiang, T.**, Wang, Z., Liang, J., Li, C., Wang, Y., & Wang, T. (2025). RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction. In *International Conference on Learning Representations (ICLR 2025)*. [pdf]
- [C7] **Jiang, T.**, Li, C., Ma, F., & Wang, T. (2025). RAPID: Retrieval Augmented Training of Differentially Private Diffusion Models. In *International Conference on Learning Representations (ICLR 2025)*. [pdf]
- [C8] **Jiang, T.**, Bendre, S. K., Lyu, H., & Luo, J. (2021). From Static to Dynamic Prediction: Wildfire Risk Assessment Based on Multiple Environmental Factors. In *2021 IEEE International Conference on Big Data (IEEE Big Data)*. [pdf]
- [C9] **Jiang, T.** & Xiong, Z. (2021). Rule-Based Approach to the Automatic Detection of Individual Tree Crowns in RGB Satellite Images. In *2021 IEEE International Conference on Computer Science, Artificial Intelligence and Electronic Engineering (IEEE-CSAIEE)*, pp. 132–135. [pdf]
- [C10] Wang, X., **Jiang, T.**, & Cai, H. II. (2021). Human Epithelial-2 Cell Image Classification Using Deep Unsupervised Learning and Gradient Boosting Trees. In *Proc. SPIE 11601, Medical Imaging 2021*. [pdf]
- [C11] **Jiang, T.**, Li, Y., Lin, H., Ruan, Y., & Woodruff, D. P. (2020). Learning-Augmented Data Stream Algorithms. In *8th International Conference on Learning Representations (ICLR 2020)*, Addis Ababa, Ethiopia. [pdf]

Journal Papers

- [J1] Tian, X., Liu, Y., Zhang, K. L., Litt, S., Shin, K., & **Jiang, T.** (2026). Do Artificial Intelligence Clients Speak Like Human Clients? Exploring GPT-4’s Content-Level Performance in Counseling Role-Play. *Journal of Counseling & Development*, 104(1), 14–26.

[J2] Tian, X., Liu, Y., Litt, S., Shin, K., & **Jiang, T.** (2025). Artificial Clients, Genuine Interactions? Assessing ChatGPT-4's Process-Level Performance in Clinical Simulations Using Qualitative Content Analysis. *Counselling and Psychotherapy Research*, 25(4), e70063.

Preprints & Under Review

[P1] Liang, J., Wang, Y., **Jiang, T.**, & Wang, T. RASA: Routing-Aware Safety Alignment for Mixture-of-Experts Models. *Under review at EMNLP 2026*. [pdf]

[P2] **Jiang, T.**, Bai, M., Pappas, N., Qi, Y., & Swamy, S. Cross-Modal Content Optimization for Steering Web Agent Preferences. *arXiv preprint*. [pdf]

[P3] Zhu, R., Wang, Y., **Jiang, T.**, Liang, J., & Wang, T. Self-Improving Model Steering. *arXiv preprint*. [pdf]

EXPERIENCE

Amazon AWS

Seattle, WA

Applied Scientist Intern, Agentic AI Lab

May 2025 – Sep 2025

- Developed cross-modal adversarial attack framework to identify vulnerabilities in VLM-based web agents, achieving up to 71% preference manipulation rate across GPT-4.1, Qwen-2.5VL, and Pixtral-Large models.
- Implemented PGD-based visual perturbation attacks exploiting CLIP encoder vulnerabilities, achieving >50% black-box transfer success while maintaining imperceptibility ($\epsilon=8/255$).
- Designed iterative textual refinement algorithms leveraging RLHF-induced biases to steer agent preferences through semantically plausible content modifications.
- Conducted comprehensive evaluation on ControlNet-100k and VisualWebArena benchmarks, demonstrating 3.5 $\times$ improvement over baseline methods while evading detection; findings released as an arXiv preprint [P2].

Stony Brook University

Stony Brook, NY

Research Assistant, Advisor: Dr. Ting Wang

Aug 2023 – Present

- Conduct research in AI safety, focusing on the safety and robustness of large language models and LLM-based agents.
- Lead multiple first-author research projects spanning LLM jailbreak defense (RobustKV, ICLR 2025), differentially private generative modeling (RAPID, ICLR 2025), adversarial robustness of vision-language models (DTR, CVPR 2026), and long-horizon attack benchmarking for LLM agents (AgentLAB, ICML 2026).
- Collaborate on research addressing memory-based defenses for LLM agents (MAGE, CCS 2026), jailbreaking of large reasoning models (AutoRAN, ACL 2026), adversarial attacks on GraphRAG systems (S&P 2026), and safety alignment for mixture-of-experts models (RASA, under review).

Pennsylvania State University

State College, PA

Research Assistant

Aug 2022 – May 2023

- Led a research project developing “Unlearnable Examples” aimed at minimizing the performance of contrastive learning models by introducing adversarial noise into training images.
- Attended and presented in weekly individual and group meetings with Dr. Ting Wang, Dr. Dongwon Lee, and lab mates.

University of Rochester, Goergen Institute for Data Science

Rochester, NY

Teaching Assistant

Jan 2021 – Dec 2021

- Held office hours averaging over 2 hours weekly to assist students with Python, Linux, SQL, PySpark, and R.
- Collaborated with Professors Lloyd Palum and Brendan Mort to test and refine course materials on Databricks.
- Conducted review sessions to reinforce course material and prepare students for assessments.
- Graded assignments and projects, providing constructive feedback to students.

HONORS AND AWARDS

Graham Endowed Fellowship (\$4,000), Pennsylvania State University, Aug 2022

NSF Student Travel Grant (\$500), IEEE Big Data Conference, Dec 2021

Silver Medal (Top 3%, 16th/541), Kaggle: Google Landmark Retrieval Challenge, 2020